PRIYANSHI

MACHINE LEARNING ENGINEER · RESEACRH INTERN · CSE-AI UNDERGRADUATE

9.5
CGPA

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- Chitkara University, Punjab

EDUCATION

B.E. — CSE (Artificial Intelligence)
Chitkara University
2024 – 2028
9.5 CGPA

Senior Secondary — CBSE
St. Mary's Convent, Panipat
2023 · 89.2% (PCMB)

SKILLS

LANGUAGES

Python · C / C++

ML & AI

PyTorch · Scikit-learn · HuggingFace · Transformers · NLP · Deep Learning

BACKEND

FastAPI · Django · Flask · REST APIs

DEVOPS & CLOUD

Docker · Nginx · Linux · Git · AWS Bedrock · S3 / EC2

FRONTEND

React · PyQt5

SCIENTIFIC

MNE-Python · NumPy · Pandas · SciPy

CURRENTLY

- **Research Intern — IIT Delhi**
  EEG · Deep Learning · NLP · Hyperspace Representations
- **Learning — Attention Mechanisms**
  Implementing 'Attention is all you need'

EXPERIENCE

Research Intern

March 2026 – Present

Indian Institute of Technology Delhi

- Conducting research on EEG-based cognitive state analysis using deep learning and NLP, focusing on information representation in high-dimensional hyperspace — including word embeddings, zero-shot and one-shot learning paradigms.
- Designing and implementing experimental ML pipelines for neural signal processing; applying MNE-Python for EEG preprocessing, artifact rejection, epoch extraction, and feature engineering for downstream classifiers.
- Collaborating directly with IIT Delhi faculty across model development, evaluation cycles, and iterative research in signal processing and representation learning.

PROJECTS

Transformer-based Web Application Firewall

2026

- Designed and trained a transformer-based classifier for real-time malicious HTTP request detection, identifying SQL injection, XSS, and path traversal attacks with 88% classification accuracy.
- Integrated the model with Nginx via `auth_request`, routing live traffic through a FastAPI inference microservice — enabling automatic blocking at the proxy layer with sub-100ms latency.
- Built the complete production pipeline: data preprocessing, PyTorch model training, Docker containerisation, and Nginx deployment on Linux.

Python · PyTorch · FastAPI · Nginx · Docker · Linux

CivicLens — AI Document Validation System

2026

- Built an AI validation pipeline using AWS Bedrock (Nova 2 Lite) to detect cross-document inconsistencies and missing fields in government application forms before submission.
- Architected a FastAPI backend and React frontend with AWS Translate for multilingual support across 5 languages, and automated document lifecycle management via S3 TTL-based deletion for compliance.
- Deployed on AWS EC2; handles multi-document uploads, parallel validation, and structured per-field error reporting.

AWS Bedrock · FastAPI · React · S3 · EC2 · AWS Translate

AI UI Navigator — Autonomous Browser Agent

2026

- Built a multimodal autonomous agent that interprets natural-language instructions and executes web tasks end-to-end using Gemini vision and reasoning with Playwright automation inside an XVFB headless environment.
- Real-time execution feedback streamed via WebSockets — agent streams step-by-step progress and completion status to a connected frontend client.
- Handles complex multi-step flows including dynamic page rendering, typing in search boxes, finding correct corrdinates to click etc.

Gemini · Playwright · WebSockets · XVFB · Python · Multimodal AI